# DB-Sorter

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## 1 Findings

- main algorithm is
$$H_{k+1} = e^{s[D,H_k]} H_k e^{-s[D,H_k]}$$
where $D$ is a diagonal with sorted diagonal elements
- Sorting characteristic of DB happens for the discrete case as well: if $c_k = tr(H_k D)$
$$c_{k+1} - c_k = tr(\|[H_k, D]\|^2) + \mathcal{O}(s^2) \geq 0 + \mathcal{O}(s^2)$$
- each step gets closer to the ground energy:
$$\frac{d}{dt}\langle 0|H(t)|0\rangle = 2\langle\psi|d_0 I - D|\psi\rangle \leq 0$$
- in general for the $k$'th excited energy level $E_k$ we can show the following which does not guarantee a result similar to above for any excited states other than the last one:
$$\frac{d}{dt}(\langle k|\, H(t)\, |k\rangle - E_k)^2 = 4(\langle k|\, H\, |k\rangle - E_k)(\langle\psi_k|\, d_k I - D\, |\psi_k\rangle)$$
- If $D$'s spectral gap is $\Delta$ and we can write $|\psi\rangle = H|0\rangle = \alpha|0\rangle + \beta|0^\perp\rangle$, then:
$$\frac{d}{dt}\langle 0|H|0\rangle \leq -2\beta^2\Delta$$
- Similar results can be achieved for the discrete case mostly with $\mathcal{O}(s^2)$ error
- for the case where $H$ has standard basis as its eigenstates, we can make $D$ diagonal in another basis
- using the above result and writing $D$ in $|v_k\rangle = H^{\otimes n}\,|k\rangle$ with only $d_0 = -\pi$ and $s = 1$ we recover grover algorithm
- we can try different $D$'s to tradeoff number of two-qubit gates and convergence speed
- if allowing degenerecies in $D$, we can show that $D$ should be a diagonal matrix with elements only 0 or 1. We can have fully random binary string for partial sort or stretch of zeros followed by ones for the rest of elements for full sort.